

Probability 2

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1 Introduction

Let Ω_X, Ω_Y be partitions of Ω and define $P_X : \Omega_X \rightarrow I, P_Y : \Omega_Y \rightarrow I$ be distributions. Take $P(X = x, Y = y) = P(x \cap y)$ for $x \in \Omega_X, y \in \Omega_Y$. Fix $x \in \Omega_X$.

$$\sum_{y \in \Omega_Y} P(x \cap y) = P\left(\bigcup_{y \in \Omega_Y} x \cap y\right) = \sum_{y \in \Omega_Y} P(x \cap \Omega) = P(x) = P_X(\{x\}).$$

Suppose $P(X = Y) = 1$.

$$\sum_{x \in \Omega_X, y \in \Omega_Y} P(x \cap y) \mathbf{1}(x = y) \leq \sum P(x \cap y) = 1,$$

equality holds iff $P(x \cap y) \neq 0 \rightarrow \mathbf{1}(x = y) = 1$. Let $x \in \Omega_X$. Consider $\sum_{y \in \Omega_Y} P(x \cap y) = P(x) > 0$, so $P(x \cap y) \neq 0$ for some $y \in \Omega_Y$. Then $x = y$ for some $y \in \Omega_Y$, or $\Omega_X \subset \Omega_Y$. Similarly, $\Omega_Y \subset \Omega_X$ and $\Omega_X = \Omega_Y$. P_X is determined by Ω_X and P , so $P_X = P_Y$.

Suppose X, Y are independent, so that for all $x \in \Omega_X, y \in \Omega_Y, P(x \cap y) = P(x)P(y)$. Then for $A \subset \Omega_X$ and $B \subset \Omega_Y$,

$$\begin{aligned} P(X \in A \cap Y \in B) &= P\left(\bigcup_{x \in A} x \cap \bigcup_{y \in B} y\right) \\ &= \sum_{x \in A, y \in B} P(x \cap y) \\ &= \sum_{x \in A, y \in B} P(x)P(y) \\ &= \sum_{x \in A} P(x) \sum_{y \in B} P(y) \\ &= P(A)P(B). \end{aligned}$$

$$P(X = x|Y = y) = P(x \cap y)/P(y) = P(x).$$

If X, Y are independent, then $E[XY] = E[X]E[Y]$.

$$\begin{aligned} E[XY] &= \sum_{x \in X, y \in Y} xyP(x \cap y) \\ &= \sum_{x \in X, y \in Y} xyP(x)P(y) \\ &= E[X]E[Y]. \end{aligned}$$

Let $X \geq 0$ be a r.v. Then for $a > 0$, $P(X \geq a) \leq E[X]/a$. Proof:

$$E[X] = \sum_x xP(X = x) \geq \sum_{x \geq a} xP(X = x) \geq a \sum_{x \geq a} P(X = x) = aP(X \geq a).$$

Let X be a r.v. and $g : X \rightarrow Y$ a function. Then $E[g(X)] = \sum_x g(x)P(X = x)$.

$$\begin{aligned} E[g(X)] &= \sum_y yP(g^{-1}(y)) = \sum_y y \sum_x \mathbf{1}(g(x) = y)P(X = x) \\ &= \sum_x \sum_y y \mathbf{1}(g(x) = y)P(X = x) = \sum_x g(x)P(X = x). \end{aligned}$$

Let X be a standardized r.v. $P(X^2 \geq k^2) \leq E[X^2]/k^2 = \text{Var}(X)/k^2 = 1/k^2$.

For independent r.v. X, Y , $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.

$$\begin{aligned} E[(X + Y)^2] - E[X + Y]^2 &= E[X^2] + 2E[XY] + E[Y^2] - E[X]^2 - 2E[X]E[Y] - E[Y]^2 \\ &= \text{Var}(X) + \text{Var}(Y) + 2(E[XY] - E[X]E[Y]) \\ &= \text{Var}(X) + \text{Var}(Y). \end{aligned}$$

If X, Y are rvs such that their supports are both cardinality 2 and $E[X]E[Y] = E[XY]$ (so X, Y are uncorrelated), then X, Y are independent.